

SOURCE

Seoul Living Migration (Saenghwal Idong), administrative-dong product
dong x weekday x arrival hour x sex x age band x trip type x volume

202001–202607 · 79 of 79 months complete · 424 dong
1,896 parquet files · 10,186,891,962 rows

WHAT THE PRODUCT DOES NOT RECORD

- no date column: a month is one row per weekday, not per day
- the volume column is the within-month sum over like weekdays (hap), not a daily mean
- cells below 3 are masked, 26.1% of cells overall, and the loss is not spread evenly:
20-24 to 40-44 lose 10.9%-69.5% of their cells, 45-64 lose under 0.13%, and 0-19 and 65+ lose exactly none
- expansion weights vary with age and are not recorded in the data
- the code space is frozen at 2020 while the districts have moved



EXTERNAL ANCHORS — what closes each gap

official district file

measures the hidden volume:
2.27 per masked cell, 11.1% of
volume, not the 1.5 midpoint

contact survey

pins the transmission
parameter; it does not
identify it

resident registration

denominators, and the coverage
profile the data does not
record



ESTIMATOR

co-arrival in the same dong, the same hour, the same weekday
→ age x age matrix A, at three spatial scales (dong 424 / district 25 / city 1)
→ excess over the proportionate-mixing null: assortativity, normalised MI, rank

THREE CHECKS, all green before any number is quoted

citation audit · independent second implementation 436/436 · bit-for-bit determinism



THREE CLAIMS

1 the instrument
responds

the co-arrival matrix tracks
the Korean school year in all
79 months, and the year schools
stayed shut is flat. The
calendar is never supplied to
it. Of the 17 months that reach
the survey's floor, every one
is a term month; 3 of 78
alternative calendar alignments
match.

Fig 2, Fig 3(c)

2 and it still reads
near zero

the departure from
proportionate mixing is near
zero at every spatial scale and
every age resolution. An
identity fixes what this
estimator can read at all, and
a recovery experiment turns
that into a bound conditional
on how much sits below the
cell.

Fig 3, Fig 4, Fig 5

3 which changes an
allocation

across the coverage family the
ranking an allocation rule
reads moves, and the SEIR cost
of reading the wrong one
depends on R_0 : largest at
1.2-1.5, and 5 of 20 verdicts
turn on it.

Fig 6, Fig 7